higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE (VOCATIONAL)

## MATHEMATICS
(First Paper)
**NQF LEVEL 4**

**5 NOVEMBER 2019**

**This marking guideline consists of 14 pages.**

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## INSTRUCTIONS TO MARKERS

1. Examiners are reminded that the subject Mathematics lends itself to a variety of problem-solving methods and therefore several different alternative answers are possible. Allocate marks for ALL mathematically correct answers.

2. The answers provided in this memorandum are not exhaustive and examiners/markers are requested to discuss all additional alternative solutions before the commencement of marking. The marking memorandum strives to minimise discretionary marking to ensure consistency.

3. Implement 'follow up' marking. In other words, allocate marks for accurate steps following an error, provided that the responses are consistent with the error. This is referred to as consistent accuracy (CA) marks. Be careful of double errors that result in an accurate answer.

4. Marks are allocated per step, but if a student omitted a step and there is evidence or reason to believe that he/she could derive the next step mentally, then the mark must still be given to the student.

5. Take note that marks have been allocated for simplification in certain questions. This is to identify high achievers.

6. All STEPS must be marked and not only the final answer. TICKS must correspond with the total.

7. If the marker comes across a novel or unusual solution please discuss it with the chief marker before proceeding.

8. Ensure that you have marked answers to every sub-question.

9. Read ALL the answers carefully.

10. Follow the answers according to the numbering system used in this question paper.

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## QUESTION 1

1.1
1.1.1 C or $-9+7i$
1.1.2 D or $-i$
1.1.3 A or $34$
1.1.4 C or Third
1.1.5 B or $x = 5$ and $y = 2$

(5 × 1) (5)

1.2
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$x = \frac{-2 \pm \sqrt{-36}}{4}\surd$$
$$x = \frac{-2 \pm 6i\surd}{4}$$
$$x = -\frac{1}{2} + \frac{3}{2}i\surd \ or \quad x = -\frac{1}{2} - \frac{3}{2}i\surd$$

½ for substitution in **correct** formula (A)
½ for conversion to complex form (CA)
½ for each answer (CA)

(2)

1.3
$$\frac{\sqrt{2}cis(-45^0)}{-2+3i+4}$$
$$= \frac{1-i\surd}{2+3i} \times \frac{2-3i}{2-3i}\surd$$
$$= \frac{2-2i-3i+3i^2\surd}{4-9i^2}$$
$$= \frac{-1-5i\surd}{13\surd} = \frac{-1}{13} - \frac{5}{13}i\surd \ or \ \ 0,08-0,39i$$

½ for convert to rectangular (A)
½ rationalise denominator (A)
½ mmultiplication in numerator (CA)
½ - simplify numerator (CA)
½ - simplify denominator (CA)
½ for answer $a \pm bi$ (2 terms) (CA)

Alternative
$$\frac{\sqrt{2}cis\left(-45^0\right)}{3,606\underline{\left|56,31\surd\right.}}$$
$$= \frac{1,414}{3,606}\surd\underline{\left|-45-56,31\surd\right.}$$
$$= 0,392\underline{\left|-101,31\surd\right.}$$
$$= -0,08\surd - 0,38i\surd$$

When rounded to two decimals the answers might differ depending from which stage the student started rounding.

(3)

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1.4 $\dfrac{(4-3i)-(i^2+i^3)}{\left(2\,cis\,120^0\right)^5}$ ½ simplify

$=\dfrac{5-2i\ \surd}{(2\,cis\,120^0)^{\,5}}$ ½ convert polar

$=\dfrac{5,39\,cis(-21,80)}{2^5\,cis\left(120^0\times 5\right)\checkmark}\surd$ 1 mark applying De Moivre

1 mark Division

$=\dfrac{5,39cis(-21,80)}{32\,cis\,600^0}\surd$ Half mark modulus
Half mark negative argument

$=0,168\,cis(-21.80-600)\checkmark$

$=0.168\surd cis\,(-621,80^0)\surd=0,168\,cis\,(98,20^0)\surd$ 1 mark Answer with positive argument (5)

1.5 $x^2i^8-y^2i^7+6xi^6-xi\ =9i^2-7i$

$x^2\surd+y^2i\surd-6x\surd-xi\ =-9\surd-7i$ Simplify $i^n$: ½ for each correct (A)

$y^2i-xi\ =-7i$

$x^2-6x-=-9$

$\therefore x^2-6x+9=0$.....1✓ 1 for correct equation 1 (CA)

$y^2-x=-7$ .....2✓ 1 for correct equation 2 (CA)

Solve eq1

$x^2-6x+9=0$

$\therefore x=\dfrac{6\pm\sqrt{36-36}}{2}$

$\therefore x=3$

or $(x-3)(x-3)=0$

$\therefore x=3\checkmark$ 1 for correct $x$ value (A)

*Substitute in* 2

$y^2-3=-7$

$\therefore y=\pm\sqrt{-4}$

$\therefore y=\pm 2i\checkmark$ ½ each for correct $y$ values (A)

(6)
[21]

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**QUESTION 2**

2.1 $f(x) = 2x^3 + px^2 + 5x + 6$

$x - 2 = 0 \therefore x = 2$

$f(2) = 2(2)^3 + p(2)^2 + 5(2) + 6 = -20$✓ ½ each for substituting &

$\therefore 16 + 4p + 16 = -20$ equating to -20 (A)

$\therefore 4p = -52$

$\therefore p = -13$✓ 1 for answer (CA) (2)

2.2 $(-2x^3 - 3x^2 + 13bx + 20) \div (x + a)$

$\therefore -2(-a)^3 - 3(-a)^2 + 13b(-a) + 20 = -56$ √ ½ for substituting $-a$ (A)

$\therefore 2a^3 - 3a^2 - 13ba + 20 = -56$

$\therefore 2a^3 - 3a^2 - 13ab = -76$ equation....1✓ 1 for equation 1 (CA)

$(-2x^3 - 3x^2 + 13bx + 20) \div (x - a)$

$-2(a)^3 - 3(a)^2 + 13b(a) + 20 = 0$ √ ½ for substituting $a$ (A)

$-2a^3 - 3a^2 + 13ba + 20 = 0$

$\therefore -2a^3 - 3a^2 + 13ab = -20$ equation...2✓ 1 for equation 2 (CA)

Equation 1 + 2

$2a^3 - 3a^2 - 13ab = -76$

$-2a^3 - 3a^2 + 13ab = -20$

$-6a^2 = -96$

$a^2 = 16 \quad \therefore a \pm = 4$✓ 1 for two $a$ values (A)

when $x = 4$

$2(4)^3 - 3(4)^2 - 13(4)b = -76a$

$\therefore -52b = -156 \quad \therefore b = 3$✓ 1 for $b$ value (A)

Alternative: when $x = -4$ on the other equation

$-2a^3 - 3a^2 - 13ab = -20$

$-2(-4)^3 - 3(-4)^2 + 13(-4)b = -20$

$b = 3$ (5)

2.3 2.3.1 $f^{-1} : y = \sqrt{-2}$

The graph does not exist at $x = 2$

OR

The graph is defined for negative $x$ values only $x \in (-\infty; 0]$ where $x \in R$ (1)

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2.3.2 $x \in (-\infty; 0]$ where $x \in R$
or $-\infty < x \leq 0$ where $x \in R$ (1)

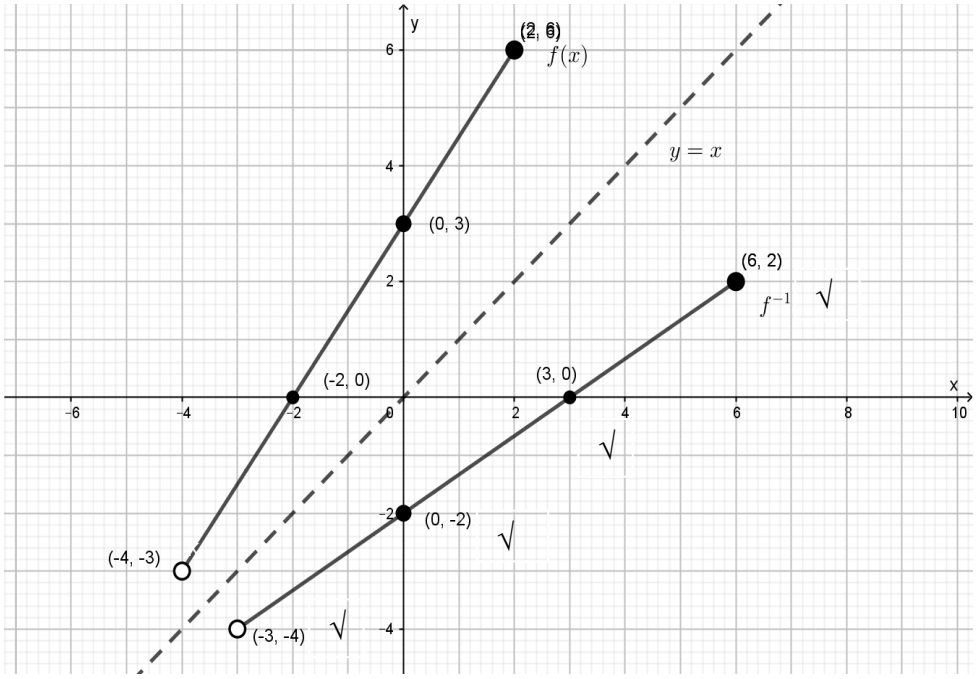
2.3.3 $y \in (-4; 2]$ where $y \in R$ (1)

2.3.4 $f^{-1}(x) = \sqrt{-x}$
$\therefore x = \sqrt{-y}$ ½ for negative (A)
$\therefore -y = x^2$
$\therefore y = -x^2$ √ ½ for $x^2$ (A) (1)

2.3.5 $g(x): y = \frac{3}{2}x + 3$
$\therefore g^{-1}(x): x = \frac{3}{2}y + 3$ 1 for interchanging $x$ and $y$ (A)
$\therefore \quad \frac{3}{2}y = x - 3$✓
$\therefore \quad y = \frac{2}{3}x - 2$ 1 for answer following interchange (CA)
$g^{-1}(x) = \frac{2}{3}x - 2$✓ (2)

2.3.6 $g(-4) = \frac{3}{2}(-4) + 3 = -3$ √
$g(2) = \frac{3}{2}(2) + 3 = 6$ √ 1 for $y$ = -3
$y \in (✓-3; 6]$✓ *where* $y \in R$ ½ for $y$ = 6
*or* $-3 < y \leq 6$ *where* $y \in R$ 1 for excluding -3
½ for including 6 (3)

2.3.7




½ for open point (-3; -4)
½ for $x$ – intercept (3; 0)
1 mark for $y = x$ line
½ for closed point (6; 2)
½ for $y$-intercept (0; -2)

(3)
**[19]**

## QUESTION 3

3.1 $f(x) = 5 - 2x^2$ Show all steps. No steps no marks!

$$f(x+h) = 5 - 2(x+h)^2$$
$$= 5 - 2(x^2 + 2xh + h^2)$$
$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$
$$= \lim_{h \to 0} \frac{5 - 2(x^2 + 2xh + h^2) - (5 - 2x^2)}{h} \surd$$ Substitution ½ mark
$$= \lim_{h \to 0} \frac{5 - 2x^2 - 4xh - 2h^2 + 2x^2 - 5}{h} \surd$$ Expanding ½ mark
$$= \lim_{h \to 0} \frac{-4xh - 2h^2 \surd}{h}$$ Simplifying numerator ½ mark
$$= \lim_{h \to 0} \frac{h(-4x - 2h)}{h}$$ Correct simplification dividing by $h$: ½ mark
$$= \lim_{h \to 0} -4x - 2h \surd$$ Correct answer 1 mark
$$= -4x \checkmark$$

(3)

3.2 3.2.1

$$f(x)=3x^2-\frac{\sqrt{x^3}}{3}+2e^{3x}-\text{cosec }4x-9x$$

$$f(x)=3x^2-\frac{x^{\frac{3}{2}}}{3}\surd+2e^{3x}-\text{cosec }4x-9x$$

Allocate ½ marks as indicated

$$f'(x)=6x\surd\ -\frac{3}{6}\surd x^{\frac{1}{2}}\surd+6^{\surd}e^{3x}\surd+4\surd\cos ec\,4x\cot 4x\surd-9\surd$$

$$f'(x)=6x-\frac{1}{2}\sqrt{x}\surd+6e^{3x}+4\cos es\,4x\cot 4x-9$$

(5)

3.2.2

$$\frac{dy}{dx}=\frac{vu'-v'u}{v^2}\surd$$

$$\frac{dy}{dx}=\frac{3\ln x(3)\surd-\frac{3}{x}\surd(3x+1)}{(3\ln x)^2\surd}$$

$$\frac{dy}{dx}=\frac{9\ln x-9-\frac{3}{x}}{(3\ln x)^2}\checkmark$$

$$or\ \frac{dy}{dx}=\frac{\left(3\ln x-3-\frac{1}{x}\right)}{3(\ln x)^2}$$

**Alternate**

$$y=(3x+1)(3\ln x)^{-1}\surd$$

$$\frac{dy}{dx}=uv'+vu'\surd$$

$$\frac{dy}{dx}=(3x+1)\left(-(3\ln x)^{-2}\right)\left(\frac{3}{x}\right)\surd+(3\ln x)^{-1}(3)\surd$$

$$\frac{dy}{dx}=\frac{9+\frac{3}{x}}{-(3\ln x)^2}\surd+\frac{3}{3\ln x}\surd$$

(3)

Allocate marks according to the ticks

3.2.3

Pr*oduct Rule*

$u=5t\ \therefore u'=5$

$v=\sin 5t\ \ v'=5\cos 5t$

$f'(t)=uv'+vu'\surd$

$\therefore f'(t)=5t(5\cos 5t)\checkmark+5\surd(\sin 5t)\surd$

$\therefore f'(t)=25t\surd(\cos 5t)+5(\sin 5t)$

Half mark RULE
$u'=5$ half mark
$v'=5\cos 5t$ 1 mark
Half mark substitution
Half mark if simplified

(3)

3.2.4 *Function of function*:

$y = \left(x^2 - 2\right)^{-2}\sqrt{}$ half mark exponential form

$\frac{dy}{dx} = -2\sqrt{}\left(x^2 - 2\right)^{-3}\sqrt{}(2x)\sqrt{}$ half mark differentiate exponent; half mark differentiate base; half mark positive exponent

$\frac{dy}{dx} = -\frac{2(2x)}{\left(x^2 - 2\right)^3} = \frac{-4x\sqrt{}}{\left(x^2 - 2\right)^3\sqrt{}}$ half mark simplify numerator; half mark positive exp.

*Chain Rule*:

*Let* $u = x^2 - 2$

$\frac{du}{dx} = 2x\sqrt{}$

$\therefore y = (u)^{-2}$ half mark $\frac{du}{dx}$

$\therefore \frac{dy}{du} = -2(u)^{-3}\sqrt{}$ half mark $\frac{dy}{du}$

$\therefore \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$ half mark Substitute chain rule

$\therefore \frac{dy}{dx} = -2\left(x^2 - 2\right)^{-3}\sqrt{}(2x)\sqrt{}$ half mark positive exponent

$\frac{dy}{dx} = -\frac{2(2x)}{\left(x^2 - 2\right)^3} = \frac{-4x\sqrt{}}{\left(x^2 - 2\right)^3\sqrt{}}$ half mark simplify numerator; half mark positive exp

(3)
[17]

**QUESTION 4**

4.1 4.1.1 $\int\left(4x + \sin 4x + \frac{4}{x} + 4\right)dx$ Allocate according to ticks

$= 2x^2\sqrt{} - \frac{\cos 4x}{4\sqrt{}}\sqrt{} + 4\ln x\sqrt{} + 4x\sqrt{} + c\sqrt{}$

(3)

4.1.2 $\int (3.\sqrt[3]{x} - \frac{3}{\sqrt{x}} + 4e^{4x} + 3\sec^2 3x)dx$

$\int (3x^{\frac{1}{3}} - 3x^{-\frac{1}{2}} + 4e^{4x} + 3\sec^2 3x)dx$ ½ each change surd to exponential (A)

$= \frac{3x^{\frac{4}{3}}}{\frac{4}{3}}\sqrt{} - \frac{3x^{\frac{1}{2}}}{\frac{1}{2}}\sqrt{} + e^{4x} + \tan 3x + c$ First two terms: ½ for each integral (A) (other three terms in next step)

$= \frac{9}{4}\sqrt[3]{x^4}\sqrt{} - 6\sqrt{x}\sqrt{} + e^{4x}\sqrt{} + \sqrt{}\tan 3x\sqrt{} + c\sqrt{}$ ½ each as indicated (CA)

(4)

4.2 $\int_{-1}^{2}(4x+3)dx$

$= 2x^2\sqrt{} + 3x\Big|_{-1}^{2}\sqrt{}$ ½ for each integral (A)

$=\left[2(2)^2+6\right]-\left[2(-1)^2-3\right]\sqrt{}$ ½ for substitutions (CA)

$=15\sqrt{}$ ½ for answer (CA) (2)

4.3 $A_1 = \int_a^b y\,dx$

$A_1 = \int_1^2 \sqrt{}(x^3-6x^2+9x-2)\,dx$ ½ for breaking into parts (A)

$A_1 = \frac{x^4}{4}\sqrt{} - \frac{6x^3}{3}\sqrt{} + \frac{9x^2}{2}\sqrt{} - 2x\Big|_1^2\sqrt{}$ ½ for each of 4 integrals (A)

$\left[\frac{(2)^4}{4}-\frac{6(2)^3}{3}+\frac{9(2)^2}{2}-2(2)\right]-\left[\frac{(1)^4}{4}-\frac{6(1)^3}{3}+\frac{9(1)^2}{2}-2(1)\right]$

$=1,25\checkmark$ 1 for answer (CA)

$A_2 = \int_a^b y\,dx$

$A_2 = \int_2^3 (x^3-6x^2+9x-2)\,dx$

$A_2 = \frac{x^4}{4} - \frac{6x^3}{3} + \frac{9x^2}{2} - 2x\Big|_2^3$

$\left[\frac{(3)^4}{4}-\frac{2(3)^3}{1}+\frac{9(3)^2}{2}-2(3)\right]-\left[\frac{(2)^4}{4}-\frac{2(2)^3}{1}+\frac{9(2)^2}{2}-2(2)\right]$

$=|-1,25|=1,25\sqrt{}$ ½ for positive area (A)

Total area $=1,25+1,25=2,5$ ✓ 1 for final answer (5)

**[14]**

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**QUESTION 5**

5.1 5.1.1 $s = -t^4 + 6t^2$

$\frac{ds}{dt} = -4t^3 + 12t$✓

½ for each correct derivative (A) (1)

5.1.2 $\frac{ds}{dt} = -4t^3 + 12t$

$\frac{d^2s}{dt^2} = -12t^2 + 12 = 0$✓

$t^2 = 1$

$t = 1\text{sec}$✓

$\therefore \frac{ds}{dt} = -4(1)^3 + 12(1) = 8m/s$✓

½ for 2nd derivative (CA)
½ for equating to zero (A)
1 for *t* value (CA)
1 for maximum value (CA) (3)

5.1.3 $s = -t^4 + 6t^2$

$\therefore s = -(1)^4 + 6(1)^2 = 5\ m$✓

½ for substituting in correct equation (A)
½ for answer (CA) (1)

5.1.4 $s = -t^4 + 6t^2$

$\frac{ds}{dt} = -4t^3 + 12t$

$\frac{d^2s}{dt^2} = -12t^2 + 12$✓

$acceleration = -12(4) + 12 = -36m/s^2$✓

1 for substituting in correct equation (A)
1 for answer (CA) (2)

5.2 5.2.1 Factor theorem & long division:

$x = 1$

$f(1) = 2(1)^3 + 3(1)^2 - 12(1) + 7 = 0$ ✓

1 for showing that f at 1 is zero (A)

$$x-1 \overline{)\ 2x^3 + 3x^2 - 12x + 7}$$

$$\begin{array}{r} 2x^2\ \surd + 5x\ \surd - 7\ \surd \\ 2x^3 - 2x^2 \\ \hline 5x^2 - 12x \\ 5x^2 - 5x \\ \hline -7x + 7 \\ -7x + 7 \\ \hline 0 \end{array}$$

½ each for 3 terms in quotient (A)

$\therefore 2x^3 + 3x^2 - 12x + 7$

$= (x-1)(2x^2 + 5x - 7)$

$= (x-1)\surd(x-1)\surd(2x+7)\surd$

½ each for 3 factors (A)

Alternative inspection

$x = 1$

$f(1) = 2(1)^3 + 3(1)^2 - 12(1) + 7 = 0$ ✓ 1 for showing that f at 1 is zero (A)

$\therefore 2x^3 + 3x^2 - 12x + 7 = (x-1)(ax^2 + bx + c)$

Inspection:

$x^3$ ____ $2x^3 = ax^3 \therefore a = 2$√ ½ each for 3 $a$, $b$ and $c$ values (A)

$x^2$ ____ $3x^2 = -ax^2 + bx^2$ and $a = 2$

$\therefore 3 = -2 + b \ \therefore b = 5$√

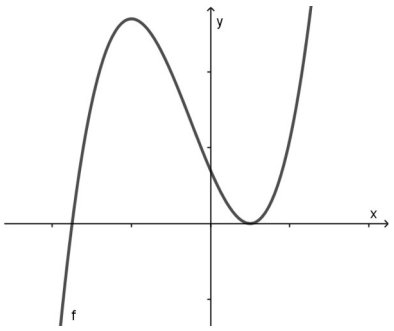
constant: $7 = -c \ \ \therefore c = -7$√

$\therefore 2x^3 + 3x^2 - 12x + 7 = (x-1)(2x^2 + 5x - 7)$ ½ each for 3 factors (A)

$= (x-1)$√$(x-1)$√$(2x+7)$√

(4)

5.2.2 A✓✓ or sketch below 2 marks answer only




(2)
**[13]**

**QUESTION 6**

6.1 6.1.1 40 pears✓ 1 for correct answer (A) (1)

6.1.2 A mix of 60 pears and 20 oranges meet the requirements of the mix.✓✓
**OR** Yes the point $(20;60)$ lies within the feasible region. (2)

6.1.3 $x + y \leq 100$√
$y \leq 40$✓;
$x \geq 10$√; ½ for each correct constraint (A)
$y \leq \frac{3}{2}x$✓

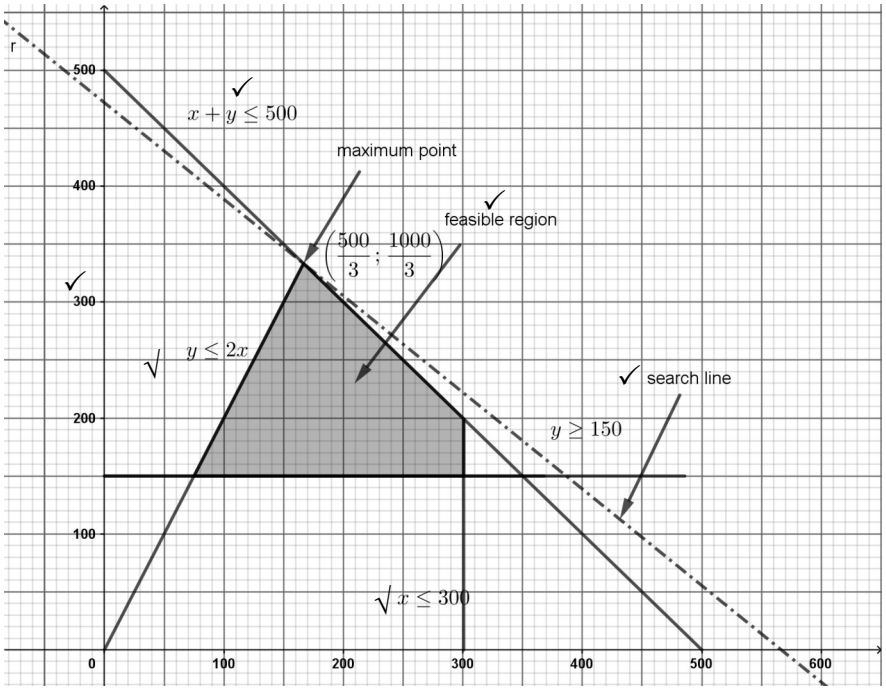
(3)

6.2 6.2.1 $x + y \leq 500\,\surd$
$y \geq 150\,\surd$
$x \leq 300\,\surd$
$y \leq 2x\,\surd$

½ for each correct constraint (A)

(2)

6.2.2 ADDENDUM B




a) 1 mark $x + y = 500$
1 mark $y = 2x$
½ for $x = 300$
½ for $y = 150$

Marks allocated for correct drawing of each line (CA)

(3)

b) Feasible region. SEE GRAPH (1)

c) $P = 250x + 300y$ (1)

d) Search line on the graph. SEE GRAPH (1)

e) Maximum profit

$y = 500 - x = 2x$

$\therefore x = \frac{500}{3}\surd$

$y = 2\left(\frac{500}{3}\right) = \frac{1000}{3}\surd$

$P = 250x + 300y$

$P = 250\left(\frac{500}{3}\right) + 300\left(\frac{1000}{3}\right)$

$P = 41666,67 + 100000 = 141666,67\checkmark$

½ each for $x$ and $y$ values (CA)
(Correctly read off from graph or calculated)

1 for max profit (CA) (2)

**[16]**

**TOTAL: 100**

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